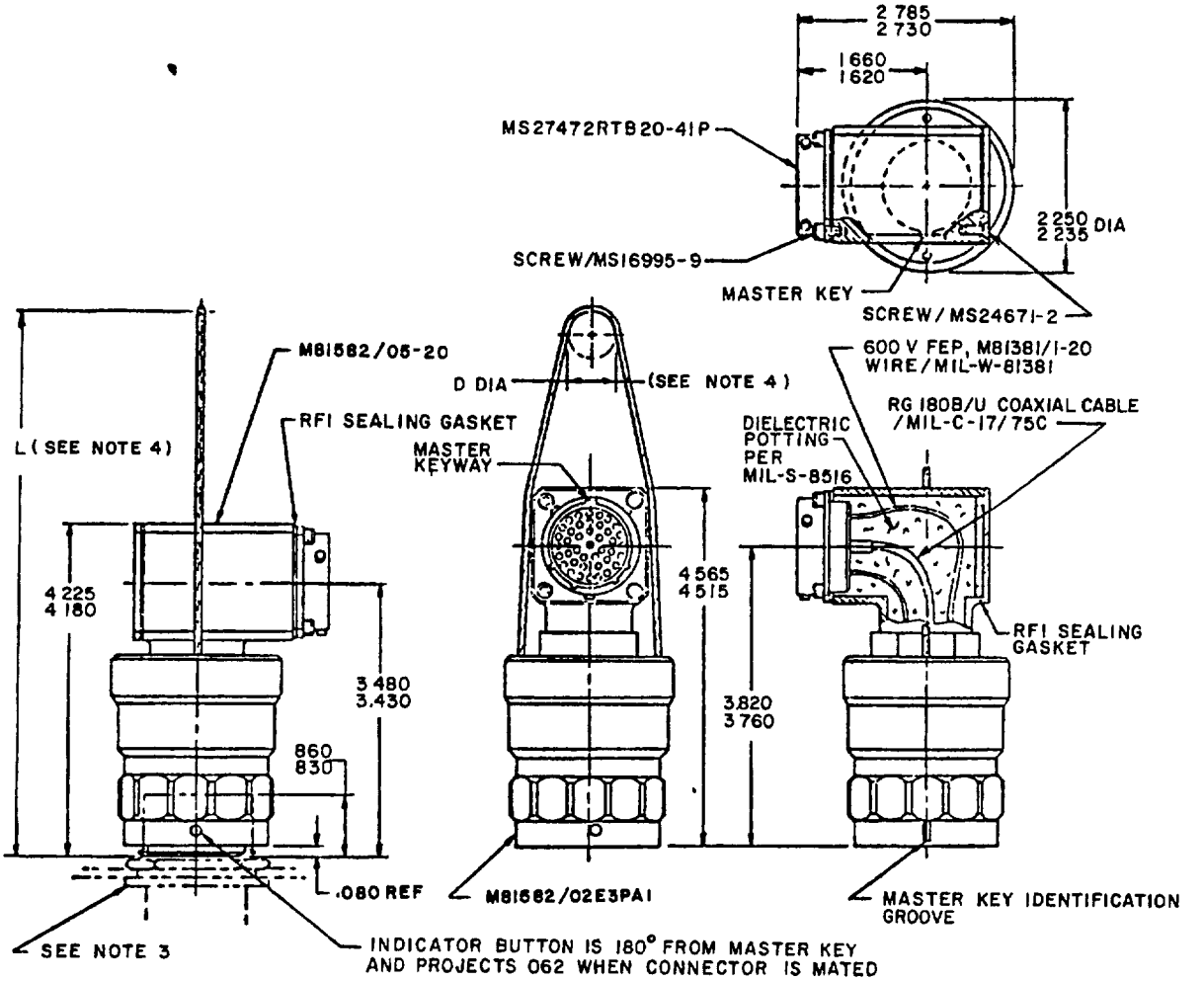
MILITARY SPECIFICATION SHEET

CONNECTOR, ADAPTER ASSEMBLY, UMBILICAL CABLE TO STORES INTERFACE, LANYARD RELEASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

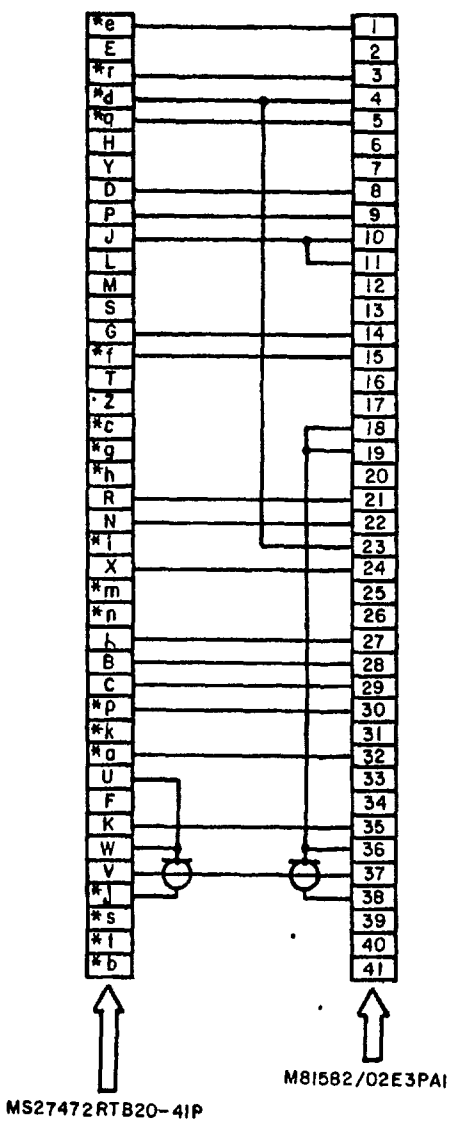
The complete requirements for procuring the adapter described herein shall consist of this document and the latest issue of Specification MIL-C-81582.



NOTES:
1. All dimensions are in inches.
2. All dimensions are after plating.
3. Mating receptacle (M81582/01E3SA) is shown to establish overall dimensions in mated condition.
4. L and D are determined by the MIL-C-81582/2 connector used.

FSC 5935

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M81582/07 adapter schematic

INCHES	MM
.050	1.27
.062	1.57
.080	2.03
.830	21.08
.860	21.84
1.620	41.15
1.660	42.16
2.235	56.77
2.250	57.15
2.730	69.34
2.785	70.74
3.430	87.12
3.480	88.39
3.760	95.50
3.820	97.03
4.180	106.17
4.225	107.32
4.515	114.68
4.565	115.95

NOTE:
1. Asterisk denotes lower case letters.

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REQUIREMENTS:

Contacts: Contacts shall conform to MIL-C-26636 or MIL-C-38999, as applicable.

Finish: The adapter finish shall be as specified, except for the MS27472 connectors which shall be plated in accordance with MIL-C-38999.

Service rating: I.

Group B inspection: Not applicable.

Qualification: Major components used in the adapter assembly shall be qualified to their respective procurement documents. Qualification testing of the adapter assembly shall be conducted on two samples in accordance with table I.

TABLE I. Sequential test schedule.

Test	Specification
Examination of product	MIL-C-81582
Dielectric withstanding voltage	MIL-C-81582/7
Circuit resistance	MIL-C-81582/7
Contact retention	MIL-C-81582/7
Temperature altitude	MIL-C-81582/7
Vibration	MIL-C-81582/7
Shock	MIL-C-81582/7
Humidity	MIL-C-81582/7
Salt spray (corrosion)	MIL-C-81582/7
Low temperature, high humidity	MIL-C-81582
Lanyard release force	MIL-C-81582
Circuit resistance	MIL-C-81582/7
Contact retention	MIL-C-81582/7
Dielectric withstanding voltage	MIL-C-81582/7
Examination of product	MIL-C-81582

Acceptance testing: Each adapter submitted for acceptance shall be subjected to the following tests:

a. Dielectric withstanding voltage as specified in MIL-C-81582/7.
b. Circuit resistance as specified in MIL-C-81582/7, and wiring conformance.
c. Contact retention as specified in MIL-C-81582.

Performance. The performance of each connector used on the adapter shall be as specified in the respective procurement documents. The adapter assembly shall perform as stated herein

a. Circuit resistance - When mated, the circuits specified on M81582/07, Adapter Schematic (sheet 2) shall not exceed the maximum millivolt drop shown in table II at a test current of 1.00 ±.05 amperes DC.

TABLE II.

Circuit	Maximum drop (millivolts)
Coax center conductor	107.8
Coax shield	17.80
Single conductor circuits	16.48
Outer shield	22.20

b. Dielectric withstanding voltage - A test voltage of 600 ±30 volts DC shall be applied between each contact and all others tied in common with the shell for 15 seconds minimum. The maximum allowable leakage shall be 25 microamperes steady state.

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c. Vibration - Vibration shall apply, except vibration cycling shall be in accordance with MIL-E-5272, Procedure XII, Curve A. Mounting shall be through the M81582/01E3SA receptacle. The MS27472 receptacle shall be mated with a compatible mating connector which shall be wired with a complete complement of wires, a minimum of one braided cable shield and cable jacket of adequate material to meet temperature requirements. The cable weight shall be .42 ounce per inch minimum. The cable length shall be 20 inches minimum and during vibration shall be clamped 8 inches minimum from the connector plug at a point external to the vibration unit.

d Shock - Mated assemblies shall be subjected to tests specified in MIL-L-81582, except shock shall be applied 3 times in each direction and test condition J shall be used.

e. Temperature - Altitude - When tested in accordance with method 504 of MIL-STD-810, Curve B for class 2 equipment, the adapters shall continuously meet the requirements for circuit resistance and dielectric withstanding voltage. Circuit resistance change shall not exceed a 28.0 percent increase

f Weight - To be added at a later date.

g. Part number: The part number shall be as shown M81582/07

Custodians
Army - EL
Navy - AS
Air Force - 80

Review activities:
Army - MU
Navy - EC
Air Force - 11, 15, 80
DSA - ES

User activity:
Navy - SH

Preparing activity:
Navy - AS

Agent
DSA - ES

(Project 5935-1953-7)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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